

[Home](#) / [Engineering](#) / [Holonomic 2D Hovercraft MPC \(MPPI\)](#)

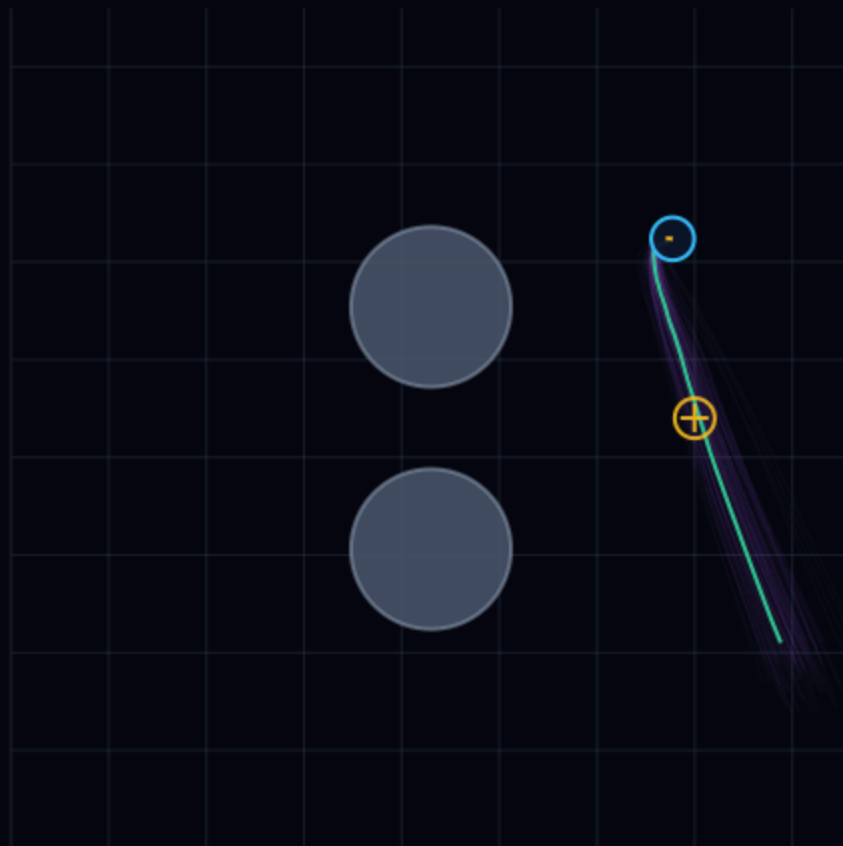
Holonomic 2D Hovercraft MPC (MPPI)

Planar double integrator with l_1 thrust cap: sampling-based MPPI steers to a draggable goal while soft-penalizing circular obstacles — rollout fan and best predicted path drawn live.

Pause

Reset

Click canvas: move goal · MPPI = weighted control averaging

$$u = (-0.26, -0.13) \text{ m/s}^2 \cdot |u|_{\max} = 1.4$$


MPPI (HOLONOMIC 2D)

Obstacle layout

Twin disks (corridor)

Horizon H

45

Samples K

84

Noise σ

0.1 m/s²

Temperature λ

4.6

Plan Δt

0.095 s

 l_{null_2} limit

1.4 m/s²

COST

Position weight

8.2

Velocity weight

0.05

Control effort R

0.21

around fixed **circular rocks** (exponential “skirt” when outside plus a strong **quadratic penalty** if the robot disk penetrates the inflated obstacle boundary). The weighted-average control update matches MPPI’s information-theoretic reweighting; the simulator applies the **first mean action** until the next replan at interval Δt . Purple fan trajectories (optional when **K** is not huge) visualize the ensemble; the **green** polyline is the **lowest-cost** sample path from the last plan. Layout presets create a **narrow corridor** or an **offset pillar field** so students can trade **horizon**, **temperature λ** , **noise σ** , and **obstacle weights** against conservatism and compute budget.

Who it's for: Students who finished the pendulum MPPI demo and want the same sampling intuition on a **2D navigation** task before touching full nonholonomic models or QP-based MPC.

KEY TERMS

- Holonomic robot
- Double integrator
- Model Predictive Path Integral (MPPI)
- Trajectory sampling
- Soft collision constraints
- Obstacle avoidance
- Warm-started control sequence

How it works

Holonomic 2D “hovercraft” with $\ddot{\mathbf{x}} = \mathbf{u}_x$, $\ddot{\mathbf{y}} = \mathbf{u}_y$ and bounded **L2** thrust: **MPPI** rolls **K** noisy acceleration trajectories over horizon **H**, adds **obstacle skirts** around circular rocks, and reweights the mean control each step — same sampling spirit as **engineering/mpc-pendulum**, now with **goal seeking** and **collision avoidance** in the plane.

Key equations

$\ddot{\mathbf{x}} = \mathbf{u}_x$, $\ddot{\mathbf{y}} = \mathbf{u}_y$, with $\|(\mathbf{u}_x, \mathbf{u}_y)\|_2 \leq u_{\text{max}}$ – holonomic double integrator (hovercraft abstraction).
MPPI: sample noisy accelerations along the horizon, score rollouts with goal, speed, effort, and obstacle soft penalties, then exponential weights update the mean control sequence.

Frequently asked questions

Why soft obstacles instead of hard “if collision discard rollout”?

Hard rejection creates **discontinuous** costs and brittle gradients for sampling methods—many rollouts die in useless configurations. Smooth penalties keep **grad-like** information in the exponential weights while still exploding cost on deep penetration.

Is this literally a hovercraft dynamics model?

Only as a **cartoon**: real hovercraft have attitude–thrust coupling, fan dynamics, and ground effect. The **double integrator with $\|\mathbf{u}\|_2$ cap** is the standard **holonomic** surrogate used in curricula to isolate **MPC in the plane**.

Why does the robot sometimes scrape obstacles despite MPC?

Soft costs allow **trade-offs**: if **q_pos** dominates or **horizon** is short, the planner may briefly accept small penetration to cut a corner. Raise **q_crash**, lengthen **H**, or increase **R** on control to be more cautious.

Does disabling the rollout fan change the controller?

No—it only hides the **purple** ensemble for performance. The **green** best-path preview and applied **u** are unchanged.

Related simulators

Continue with similar topics in this category — or [all 48 in Engineering](#).

[View category →](#)

NEW **UNIVERSITY / RESEARCH**

MPC Pendulum Swing-Up (MPPI)

▶ [Launch Simulator](#)

Sampling-based Model Predictive Control: K candidate torque rollouts over horizon H, MPPI cost-weighted...

NEW **SCHOOL**

RRT Path Planner (grid)

▶ [Launch Simulator](#)

Same 40×28 wall map as A*: random samples, nearest-neighbor steer, goal bias, collision-checked edges; gro...

NEW **UNIVERSITY / RESEARCH**

Cart–Pole: LQR vs MPC vs PID

▶ [Launch Simulator](#)

Nonlinear cart inverted pendulum: linearize about upright $\varphi = \theta$, discrete-time LQR from DARE ($Q, R \rightarrow K$),...

NEW **UNIVERSITY / RESEARCH**

Particle Filter Localization (MCL)

▶ [Launch Simulator](#)

1D or 2D toy robot with fixed range beacons: predict with noisy odometry, update with Gaussian range...

NEW **UNIVERSITY / RESEARCH**

EKF SLAM (toy)

▶ [Launch Simulator](#)

Augmented state $[x,y,\theta, \text{landmarks...}]$: noisy planar odometry predict, range & bearing updates with known...

NEW **UNIVERSITY / RESEARCH**

ADC / DAC (Sampling)

▶ [Launch Simulator](#)

Sine $\rightarrow$ samples $\rightarrow$ quantization $\rightarrow$ ZOH; Nyquist and optional aliasing demo.